

Kaggle Competition Report

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 - **Section:** BAI-4A
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1. Competition Overview

- **Competition Name:** Predicting Irrigation Need (Playground Series - Season 6, Episode 4)
 - **Problem Type:** Multiclass Classification (Target: Low, Medium, High)
 - **Evaluation Metric:** Balanced Accuracy
 - **Brief understanding of the dataset:** The dataset contains various environmental and soil-related features such as Soil pH, Moisture, Temperature, Humidity, and Rainfall to determine the irrigation requirements for different crop types.
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2. Data Preprocessing

- **Missing values handling:** Evaluated the dataset for null entries; since the data was largely complete, standard imputation or row-checks were used.
 - **Encoding techniques:** Utilized **Label Encoding** to transform categorical text features (like Soil_Type, Crop_Type, and Region) and the target variable (Irrigation_Need) into numerical format for model compatibility.
 - **Feature scaling:** Applied **StandardScaler** to normalize the numerical features, ensuring that variables with large ranges (like Electrical_Conductivity) did not disproportionately influence the models.
 - **Train-validation split:** Used an 80-20 split for initial validation before moving to full cross-validation.
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3. Models Attempted

Model	Accuracy Obtained
Naive Bayes	82.58%
Logistic Regression	74.89%
Decision Tree	98.48%
Random Forest (Final)	98.51%

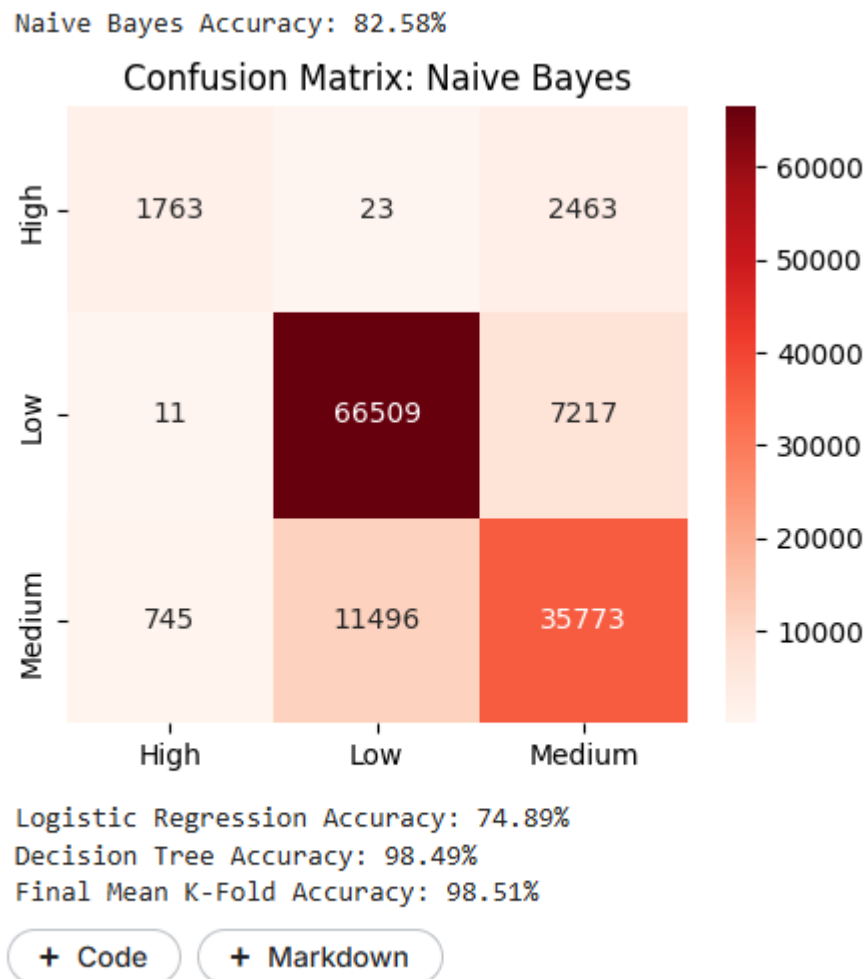
4. Cross Validation Strategy

- **K-Fold details:** Implemented **3-Fold K-Fold Cross-Validation** to ensure the model's performance was consistent across different subsets of the 630,000 rows.
 - **LOOCV observations:** Observed that Leave-One-Out Cross-Validation was computationally prohibitive for a dataset of this size; K-Fold provided the best balance of speed and reliability.
 - **Best validation accuracy:** 98.51%.
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5. Failed Attempts and Insights

- **Model used:** Naive Bayes
- **Accuracy obtained:** 82.58%
- **What went wrong?** Naive Bayes assumes that all input features are independent of each other. In this agricultural dataset, features like Humidity and Soil_Moisture are naturally correlated, leading to suboptimal classification.

- **What was improved?** Moved to **Ensemble Learning (Random Forest)** to allow the model to capture non-linear relationships and interactions between multiple soil and weather features.
- **Screenshot :**



6. Final Model Selection

- **Best model:** Random Forest Classifier
- **Hyperparameters:** `n_estimators=150`, `max_depth=15`, `random_state=7`
- **Why selected?** It yielded the highest local validation score and handled the mixture of categorical and numerical data more robustly than single-tree or linear models.

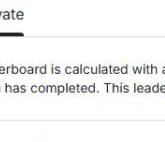
7. Leaderboard Performance

- **Kaggle score:** 0.95817
- **Rank:** Late Submission (Completed after the official prize window closed)
- **Screenshot :**


KAGGLE · PLAYGROUND PREDICTION COMPETITION · 4 HOURS AGO



Late Submission



Predicting Irrigation Need

Playground Series - Season 6 Episode 4

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YOUR RECENT SUBMISSION



submission.csv

Submitted by SyedHanzalaAli · Submitted 6 minutes ago

Score: 0.95817

Public score: 0.95584

↓

Jump to your leaderboard position



Public

Private

The private leaderboard is calculated with approximately 80% of the test data.
This competition has completed. This leaderboard reflects the final standings.

#	▲	Team	Members	Score	Entries	Last	Solution
1	▲ 73	cstdy		0.98158	128	5h	

8. Conclusion and Learnings

- **Key insights:** High-capacity ensemble models like Random Forest are far superior to simpler models when dealing with large, complex datasets containing correlated environmental features.
- **Challenges faced:** Managing a dataset with over 600,000 rows requires efficient memory management and careful scaling.

- **Future improvements:** Implementation of Gradient Boosting Machines (like XGBoost or CatBoost) could potentially push the accuracy closer to 99% by focusing on the misclassified "Medium" irrigation cases.